GGG-W-648a
November 25, 1966

SUPERSEDING
Int. Fed. Spec. GGG-W-00648 (GSA-FSS)
January 18, 1965 and
Fed. Spec. GGG-W-644
June 23, 1961 (In part) (See 6.3)

FEDERAL SPECIFICATION

WRENCH, LINEMAN'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lineman's wrenches used in pole line work. The wrenches were formerly covered by GGG-W-644.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and sizes.* Wrenches shall be of the lineman's type and of the following sizes, as specified (see 6.2)

5/8—(See table I).
3/4—(See table I).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ-T-580—Tool Steel, Carbon and Carbon-Vanadium.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

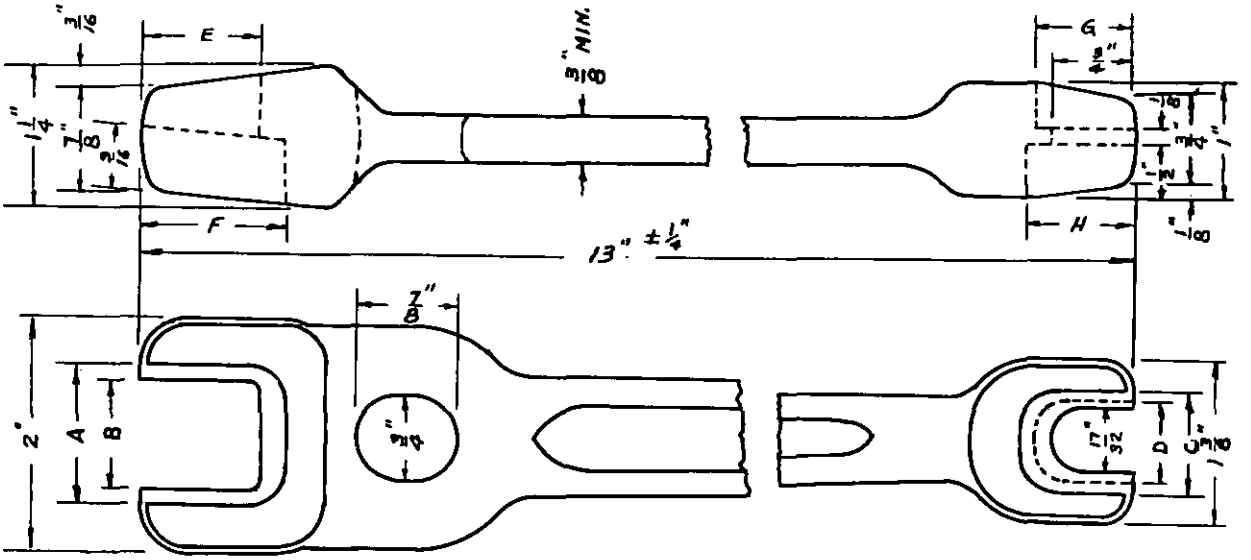
3. REQUIREMENTS

3.1 Illustration. Except for dimensional requirements, the illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of wrenches which are otherwise in accordance with the requirements of this specification.

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3.2 **Material.** Material shall be free from all defects and imperfections that might affect the serviceability of the finished product. The steel shall be in accordance with QQ–T–580, grade C, class W1–08.

3.3 **Design.** The wrenches shall be of the double open-end type and shall be of forged steel with opening surfaces having a Rockwell hardness number of not less than 35 nor more than 50 on the "C" scale. The forgings shall be in accordance with good commercial practice. The width of each wrench opening (see A, B, C, and D on fig. 1) shall be of the sizes specified in table I, as applicable. The wrench shall have a hole in the shank near the larger end for turning standard pole steps. The wrench shall withstand a test load of 4,500 inch-pounds with the pole step turning hole engaged and the load applied near the end of the small head. Each wrench opening shall withstand a test load of 7,500 inch-pounds applied near the end of the wrench opposite the head being tested. The wrenches shall be similar to figure 1.



NOTE: A tolerance of plus or minus 10 percent will be permitted on non toleranced dimensions.

Figure 1. Lineman's wrench

Table I

Size	A +1/64	B +1/64	C +1/64	D +1/64	E	F	G	H
	- 0	- 0	- 0	- 0	±1/32	±1/32	±1/32	±1/32
No.	*Inches*	*Inches*	*Inch*	*Inch*	*Inches*	*Inches*	*Inch*	*Inches*
5/8	1 3/32	29/32	13/16	5/8	1	1 1/4	13/16	7/8
3/4	1 5/16	1 1/16	7/8	5/8	1 1/8	1 13/32	27/32	1 1/32

3.4 **Finishing.** Wrenches shall be finished in accordance with the manufacturer's standard practice.

3.5 **Identification marking.** Each wrench shall be marked in a plain and permanent manner with the name or trademark of the manufacturer so that the source of manufacture may be readily determined.

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3.6 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice. Burrs, sharp edges, forging flash, and other harmful or extraneous material shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling for conformance. Sampling shall be performed in accordance with MIL-STD-105. The sample unit shall be one wrench. Data for sampling shall be as stated in table II.

Table II. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level (AQL)	AQL expressed in terms of	Reference
Visual examination	One wrench	II	4.0	Defects per 100 units	4.3.1
Dimensional examination	One wrench	S-4	1.0	Defects per 100 units	4.3.2
Testing	One wrench	S-3	1.0	Defects per 100 units	4.4
Examination of preparation for delivery	One container	II	4.0	Defects per 100 units	4.5

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table III.

Table III. *Defects list*

Examine	Defects
Material	Not as specified.
Finish	Not manufacturers' standard practice.
Design	Not as specified.
Workmanship	Burrs, sharp edges, forging flash, or other extraneous material.
	Not highest commercial grade.
Marking	Missing, incomplete, illegible, or not permanent.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 *Load test.*

4.4.1.1 *Preparation.* Straight parallel lines shall be scribed the full length of the top and bottom sides of the wrench. It is suggested that the sample wrench be securely fixed in a definite position on a vertical faceplate which is attached to a horizontal faceplate, and the straight lines scribed by means of a surface gage. The internal dimension of the wrench openings shall be measured. Suitable mandrels shall be supplied to fit into each of the wrench openings to provide

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reaction against the load. All mandrels shall be hardened to a Rockwell hardness not less than C55.

4.4.1.2 *Application of load.* Wherever possible, the load shall be applied near the end of the handle, or as far away as practicable from the wrench opening or slot. A shallow V-groove shall be filed on the wrench handle to mark the point of load application. The load shall be applied through a knife-edge bearing fitted into the V-groove and shall be held for one minute. The load shall be applied with suitable torque-producing machines or by means of dead weights or testing machines. Care shall be exercised to maintain accuracy in applying the load and in measuring the effective lever arm through which the load acts. The effective lever arm, when the load is vertical and a lever is employed, is the shortest distance in inches between two vertical lines passing respectively through the point of load application and the center of the reaction mandrel. The specified test load shall be the product of the effective lever arm, in inches, and the applied load, in pounds.

4.4.1.3 *Determination of test results.* After application of the load test, the test wrenches shall be replaced in the same position on the vertical base plate. The scriber point on the surface gage shall be moved along the scribed lines on the wrenches. Any permanent deformation of handles will be indicated by a change in the straight reference line scribed on the wrench before application of load. Slots and wrench openings shall be remeasured and examined. Any evidence of permanent set in the handles; distortion or spreading of the slots and openings; spalling (other than a brightening of the metal) of slots, or other indications of weakness of any nature that would affect serviceability of the wrench shall be cause for rejection.

4.4.2 *Hardness test.* A hardness test shall be conducted in accordance with Fed. Test Method Std. No. 151 to determine compliance with 3.3.

4.5 **Examination of preparation for delivery.** Inspection to determine compliance with preparation for delivery shall be made in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, packing, and marking.** Unless otherwise specified (see 6.2), the wrenches shall be preserved, packaged, and packed in accordance with MIL–H–15424 (as modified by 5.1.1), as specified for the applicable levels of preservation, packaging, and packing.

5.1.1 As part of the requirements of section 5 of this specification, table IV is considered as part of table I of MIL–H–15424:

Table IV

Col. 1	*Col. 2*	*Col. 3*	*Col. 4*
Wrench: Lineman's	P–1	I	72

5.1.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A in MIL–H–15424.

5.1.1.2 *Level C.* Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. (Vendor to user.)

5.1.2 *Packing.*

5.1.2.1 *Level A.* Level A packing shall be in accordance with level A packing in MIL–H–15424.

5.1.2.2 *Level B.* Level B packing shall be in accordance with level B packing in MIL–H–15424.

5.1.2.3 *Level C.* Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers

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shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The size 5/8 wrench is intended for use on 5/8-inch hardware, and the size 3/4 wrench for use on 3/4-inch hardware, in pole linework.

6.2 Odering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 Supersession data. This specification covers lineman's wrenches, formerly type III, in GGG-W-644.

Military Coordinating Activity:

YD

MILITARY CUSTODIANS:

Army—GL

Navy—YD

Air Force—69

Military review activities:

Army—GL, EL

Navy—YD

Air Force—69

Military user activities:

Army—MO, WC

Preparing activity:

GSA-FSS

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